Course Name: Elements of Environmental Science

Course Code: Env 103

Assignment-01

Semester: Fall 2026

Assignment Topic: "The Changing Atmosphere: Natural Evolution and Human Impact"

Assignment Details:

Please prepare your assignment on the topic above and follow the instructions below carefully. No excuses will be accepted for failing to follow the guidelines.

1. Introduction

- Briefly define the Earth's atmosphere and explain its importance for life, climate, and protection from solar radiation.
- Outline the present composition and layered structure of the atmosphere (troposphere, stratosphere, mesosphere, thermosphere, exosphere).
- Introduce the idea that the atmosphere has changed in two distinct ways: naturally over geological time (natural evolution) and through human activity in recent history (human impact).

2. Natural Evolution of the Atmosphere

- Explain how the atmosphere has evolved from the early Earth to the present.
- Discuss the role of biological processes such as photosynthesis in shaping the modern atmosphere.
- Explain how the layered structure and the ozone layer developed, and how they function.
- Use a simple timeline or diagram to illustrate the major stages of change.

3. Human Impact on the Atmosphere

- Analyze how human activities (fossil fuel combustion, deforestation, agriculture, industrial emissions) are altering atmospheric composition.

- Use real-world examples, case studies, or regions (e.g., rising Carbon dioxide around the world, ozone depletion over Antarctica, air pollution in South Asia).
- Mention available data, reports, or estimates (e.g., IPCC reports, NOAA, NASA, WHO, or other renowned government or NGO data) to support your analysis.
- Compare the pace of natural atmospheric change with the pace of human-driven change.

4. Conclusion

- Summarize the key points discussed.
- Reflect on the broader importance of understanding how the atmosphere has changed.
- End with a thoughtful closing statement on protecting the atmosphere going forward.

Note on Visuals

Please include 1-2 relevant pictures/graphs to support and illustrate your content (e.g., a diagram of atmospheric layers, a graph of atmospheric Carbon dioxide over time, a timeline of atmospheric evolution). Mandatory.

Marks Rubric (Total = 05 Marks)

Component	**Marks**
Introduction	0.5
Natural Evolution of the Atmosphere	1.5
Human Impact on the Atmosphere	1.5
Conclusion	0.5
Referencing + Citations	0.5
Pictures	0.5

Submission Guidelines

1. **Cover page:** Include the assignment title, course title, your name, and ID.
2. **Main Body:** Write the title first, then start your assignment. Include headings.
3. **Word limit:** 500 words (±50). Deviations may result in penalties. References, citations, and image captions are not included in the word limit.

4. **Font:** Times New Roman, size 12, 1.15 line spacing.
5. **File format:** Submit as .doc or .docx.
6. **Plagiarism:** Strictly prohibited. AI-generated content, copying from the internet, or fellow students will result in penalties. Pictures simply copied and pasted from Google will also be penalized. Use proper references for any images and graphs. AI and a similarity index over 30% are not allowed.
7. **Referencing:** APA 7th style must be followed for images, data, and quotes.
8. **File naming format:** StudentID_Name_Section (Example: 100003542_Adeela_4)
9. **Submission:** Submit to your section's Turnitin class only. Submissions elsewhere will incur penalties. The Turnitin Class ID and Enrollment Key will be provided in a separate post.

Final Reminder

Before submitting:

- Follow the instructions thoroughly and strictly.
- Double-check your word count, file format, and referencing.
- Make sure pictures are original or cited.
- Submission Deadline: Before **11:59 PM, 31st October**